Exponential Growth & Decay

Exam code: 9MA0

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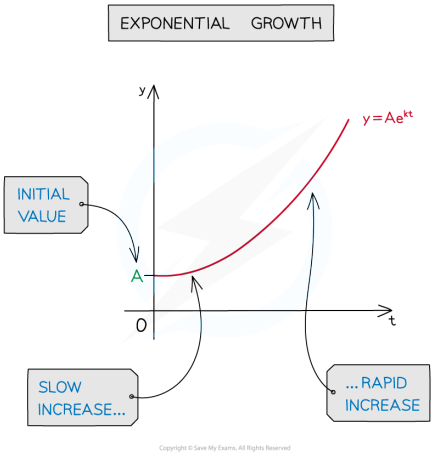
Exponential growth & decay

What is exponential growth?

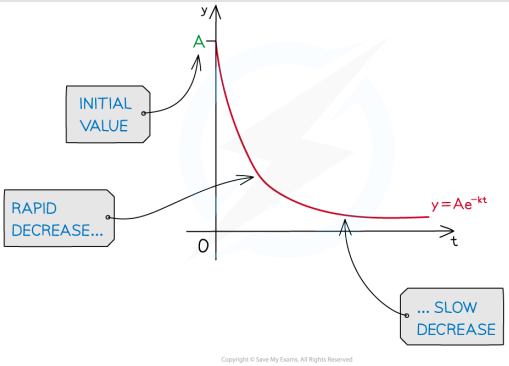
- $y = Ae^{kt}$ is exponential **growth**



What is exponential decay?

- $y = Ae^{-kt}$ is exponential **decay**

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it



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- **k > 0** so plus or minus sign in equation immediately indicates growth or decay
- **A** is the **starting** value or **initial** value, when **t = 0**
- **t** is often used, rather than **x**, as many models are time-dependent

How do I write a^x in terms of e^x?

- Using **e** makes the maths easier
- With the proper choice of **k**, all exponential functions of the form $\mathbf{a^x}$ can be written in the form $\mathbf{e^{kx}}$
  - $\mathbf{a^x = e^{kx}}$

e.g. WRITE $\left(\frac{1}{5}\right)^x$ IN THE FORM e^{kx}

$\left(\frac{1}{5}\right)^x = e^{-kx}$

$(5^{-1})^x = e^{-kx}$ LAWS OF INDICES

$5^{-x} = e^{-kx}$

$\ln 5^{-x} = \ln e^{-kx}$ LAWS OF LOGARITHMS

$-\cancel{x}\ln 5 = -k\cancel{x}\overset{1}{\cancel{\ln e}}$ $\ln e = 1$

$k = \ln 5$

$\therefore \left(\frac{1}{5}\right)^x = e^{-x\ln 5}$

NOTE: THIS WOULD BE EXPONENTIAL DECAY

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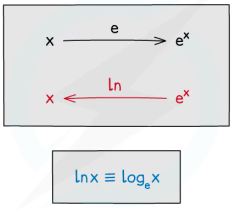
How are e^x and ln x inverses of each other?



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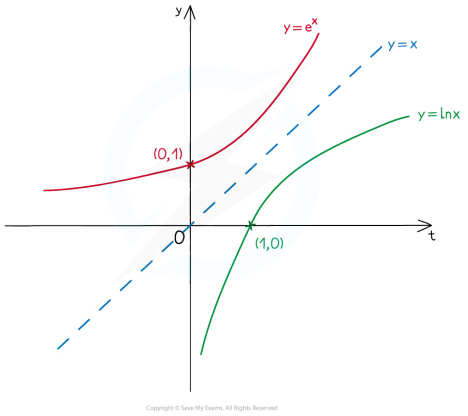
GRAPH OF y=lnx



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- The graph of **y = ln x** is the reflection of the graph of **y = e^x** in the line **y = x**

Worked Example

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